

5	(a)	Peak value is the amplitude of the a.c. signal. Root mean square value of an alternating current is equivalent to the <u>steady direct current</u> which dissipates <u>heat</u> at the <u>same average rate</u> in a given resistor.		B1
	(b)	$V_{rms} = \sqrt{\langle V^2 \rangle} = \sqrt{\frac{1}{2} \left(\frac{V_0^2}{2} \right)}$ $= \frac{V_0}{2}$ Accept solution by drawing		C1 A1
	(c)	(i)	$P_{loss} = I^2 R = \left(\frac{20.0 \times 10^6}{240 \times 10^3} \right)^2 \times 20.0 \times 10$ $= 1.39 \times 10^6 \text{ W}$	M1 A1
		(ii)	$P_{loss} = I^2 R = \left(\frac{20.0 \times 10^6}{16 \times 10^3} \right)^2 \times 20.0 \times 10$ $= 3.125 \times 10^8 \text{ W}$ Money saved = $(3.125 \times 10^8 - 1.39 \times 10^6) \times 10^{-3} \times 24 \times 0.10$ $= \$747\,000$	C1 C1 A1